

Text Classification in NLP

CMPE346 · Group 2

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Group task

Text Classification

Metric

Accuracy & Weighted F1

Student	Task
Atakan Gül	Media Bias Detection
[Student 2]	[Topic]
[Student 3]	[Topic]

BiLSTM

Bidirectional LSTM trained from scratch. Captures sequential context in both directions.

BERT `bert-base-uncased`

Transformer encoder pretrained via masked language modeling. Fine-tuned end-to-end.

RoBERTa `roberta-base`

Optimized BERT: removes next sentence prediction, uses dynamic masking and more data.

Task

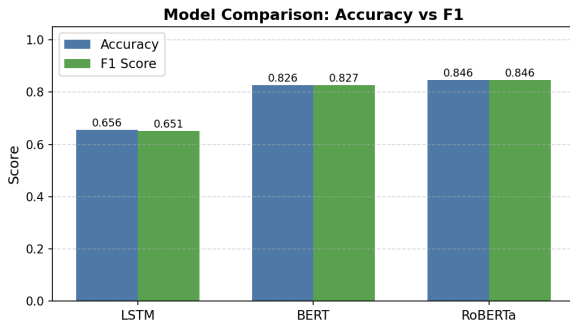
Classify a news sentence as *biased* or *not biased*.

Dataset: BABE

- 4,121 expert-annotated sentences
- Diverse news outlets
- 3,121 train / 1,000 test

Model	Acc.	F1
BiLSTM	0.656	0.651
BERT	0.826	0.827
RoBERTa	0.846	0.846

- RoBERTa best: +19% over BiLSTM
- BERT close behind
- BiLSTM struggles on subtle bias



- Media bias is expressed through framing and tone — not facts
- Pretraining is critical: transformers capture patterns BiLSTM cannot
- RoBERTa's optimized pretraining gives it a clear edge over BERT

Task

[Description]

Dataset: [Name]

Stats

Train / test split

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure]

ding 1

ding 2

inding 1

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inding 3

Task

[Description]

Dataset: [Name]

Stats

Train / test split

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure]

ding 1

ding 2

inding 1

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Task	Model	Acc.	F1
Media Bias	BiLSTM	0.656	0.651
	BERT	0.826	0.827
	RoBERTa	0.846	0.846
[Student 2]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—
[Student 3]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—

- RoBERTa consistently outperforms across all three tasks
- BERT and RoBERTa adapt well regardless of classification domain
- Pretraining is decisive — BiLSTM cannot compete on small datasets